IGNITE EDGE

PRESENTS

A PLUS CHARGE

CATALYSIS'T IIT GUWAHATI

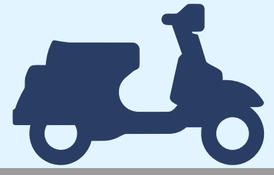
BECAUSE EVERY JOURNEY DESERVES A SMOOTH CHARGE



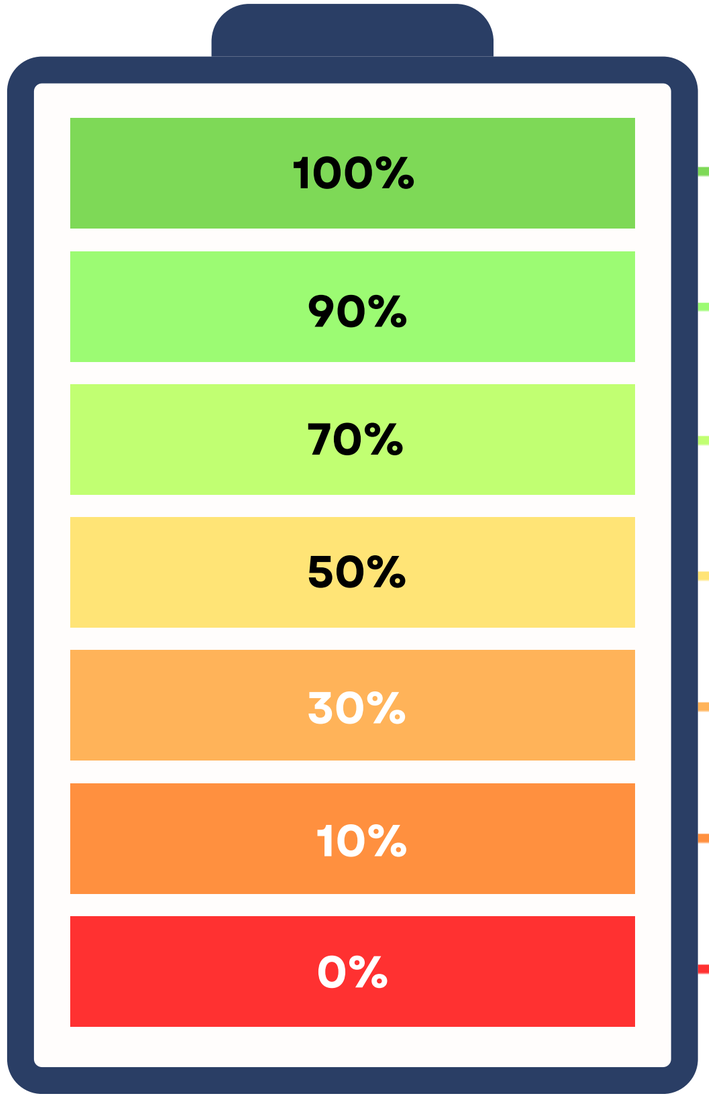








AGENDA



- CHALLENGES AND LEARNINGS
- UX DESIGN - SOFTWARE AND HARDWARE
- BUSINESS DEVELOPMENT PLAN
- WEBSITE AND SOCIAL MEDIA
- OUR UNIQUE CHARGING FRAMEWORK
- INDUSTRY LANDSCAPE AND GAP ANALYSIS
- PAIN POINTS IN EV CHARGING

PROBLEMS FACED BY THE USERS

Current Pain Points in EV Charging

Power Instability in Rural Areas

Impacts reliability in Tier 3 towns

High Wait Times & Queues

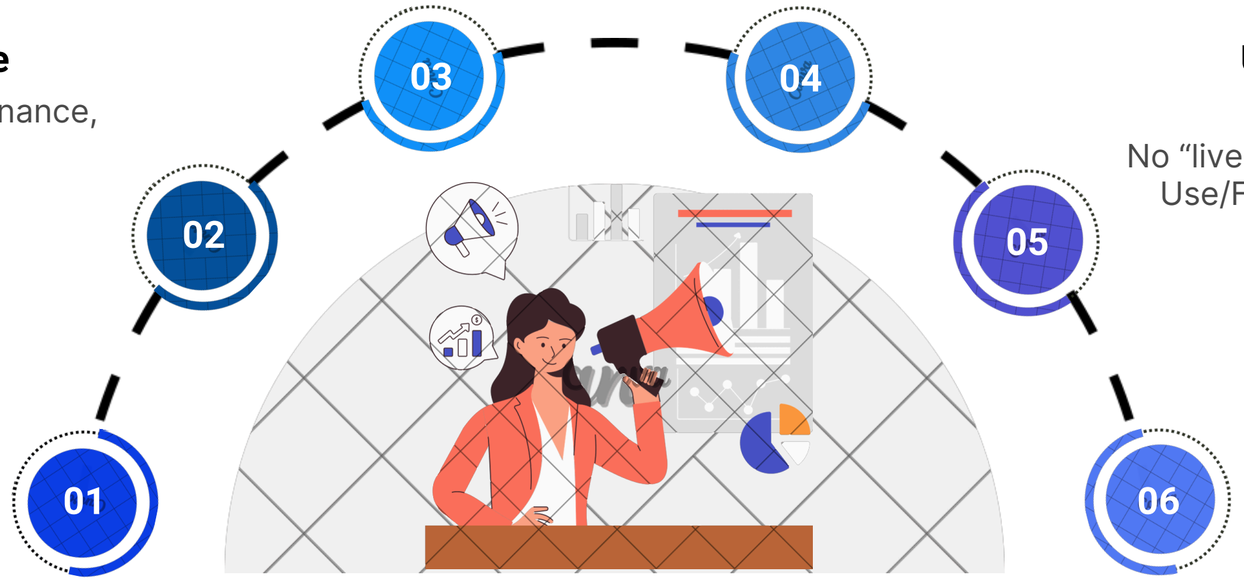
No booking/reservation system

Frequent Downtime

No predictive maintenance, fault alerts missing

Unreliable Charger Availability

No “live status” (Available/In Use/Faulty) → failed visits



Unreliable Charger Availability

No live status (Available/In Use/Faulty) → failed visits

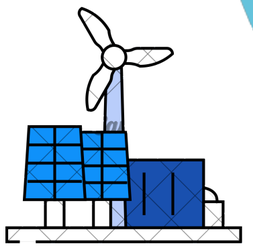
User Experience Gaps

Lack of clear instructions, low-light visibility, accessibility issues

INDUSTRY LANDSCAPE AND GAP ANALYSIS



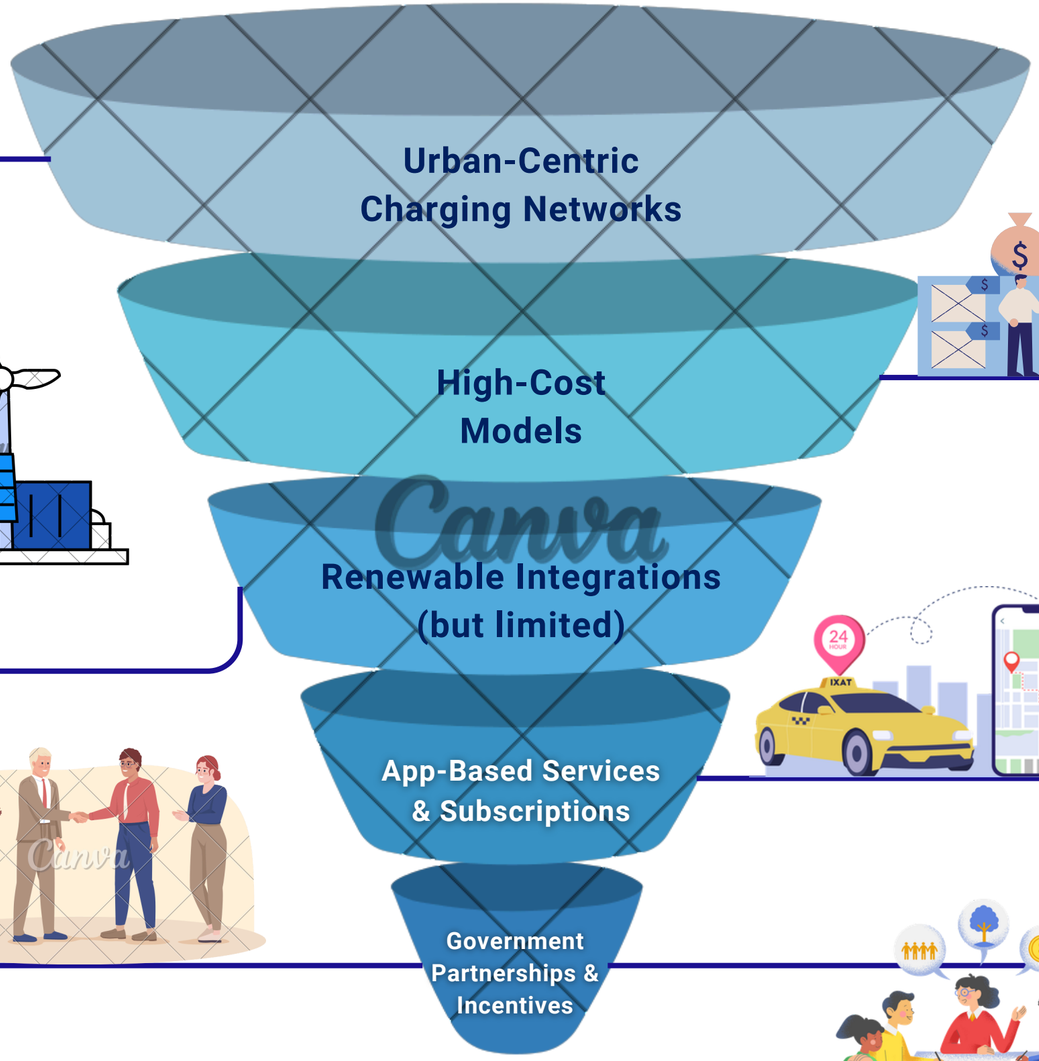
- Most big companies (Tata Power EV, Statiq, Fortum, Charge+Zone) focus mainly on **metro cities & highways**.
- Example: Tata Power → 50,000+ chargers, mostly in urban malls, housing complexes, and corporate hubs.
- Gap: Tier-2/Tier-3 cities & remote regions (like North-East) are **still underserved**.



- A few are experimenting with **solar-powered charging hubs** (like Magenta, SunFuel).
- However, renewable integration is still limited → usually pilot projects, not mass adoption.



- Companies like Fortum and Tata Power partner with state governments and PSUs to roll out chargers.



- Private charging operators invest in fast-charging DC stations (15–60 mins charging time).
- But these require huge CAPEX + land + power infra, making them expensive to deploy in less-developed regions.



- Many players (Statiq, Jio-bp Pulse) offer **apps for locating chargers, booking slots, digital payments**.
- They also test **subscription packages** (monthly charging plans) → but these work well only where charging density is high.



- Example: Indian Oil + NTPC collab to set up EV chargers at fuel stations.
- Still, implementation is slow in remote & hilly terrains (like NE states).







⚡ U-CHARGE: END-TO-END EV CHARGING FRAMEWORK

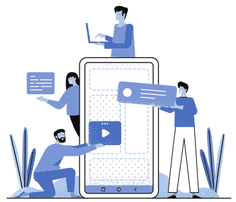
PLUS CHARGE

U C H A R G E

01

Unified AI Feedback loop

- Taking feedback from regular customers after identifying them.
- Agentic AI based automatic calling after charging



02



Community Hubs & Awareness building

- Companies + local outreach to educate people in community in which there are no EVs but has potential to have public OR Residential chargers

03

Hassle-Free Charge Simplicity

- Auto retractable chargers in malls, offices and complexes to let other vehicle charge when one vehicle is connected to charger even after fully charged



04



Adaptive Pricing & Balanced Load

- Dynamic Pricing technique i.e. charging price will be high when demand is high and vice versa.
- Load shift to reduce grid stress.

05

Rapid-Response Fast Chargers

30-Min Highway Supercharge

- 150–350 kW Level 3 DC Chargers
- Premium rate for time-sensitive users
- Late breakeven period but on longer run, will become very profitable



06



Government/ Public Spaces

Powering Public Mobility

- Public charging incentives & subsidies
- Corporate credits for employees

07

Efficient AI Guided Session Planning

AI Planning, Zero Anxiety

- “Best time to charge” alerts
- AR-based charger guidance





WEBSITE AND SOCIAL MEDIA SUGGESTION

Website

Positioning:

Central hub for subscription-based home EV charging with flat pricing, convenience, and transparency.

Content:

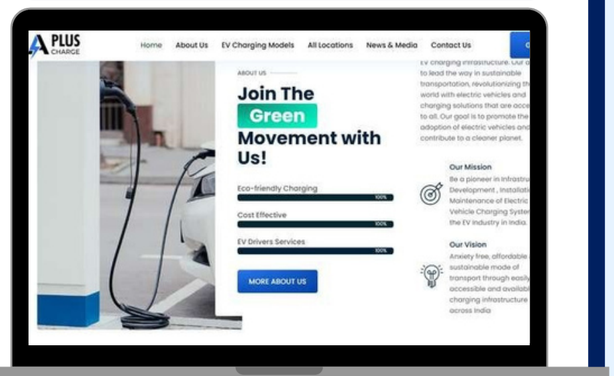
Emphasizes no installation/maintenance costs, universal compatibility, and customer focus.

SEO & Accessibility:

Optimized for EV charging/subscription keywords; mobile-friendly but SEO metrics not public.

Weaknesses:

Missing company background, team info, and charger specs; adding blogs, case studies, or testimonials could boost credibility.





Instagram

Positioning:

Visual showcase of installations, testimonials, and EV education; highlights sleek design and sustainability.

Engagement:

Uses Reels/Stories for higher reach (avg. ~1.16% engagement); Reels stress easy installs & savings.

Challenges:

Handle visibility unclear; possible low posting consistency. Misses trending formats that drive reach.

LinkedIn

Positioning:

Showcases installations, sustainability, team culture, and founder insights—building credibility as an EV-infra player.

Engagement:

Steady likes/comments; updates on projects, events, and hiring. Leadership posts add thought leadership.

Opportunities:

Stronger B2B presence via case studies, technical/market articles, LinkedIn Groups, and targeted campaigns.

Weaknesses:

Small follower base, content lacks strategy; more data-driven storytelling and long-form updates needed.



Campaign Highlight

"Charge, Snap & Share – #NEGreenCharge"

Encourage users to post selfies at charger sites or scenic NE landmarks with the campaign hashtag.

01



Reward: ₹20 cashback credited in-app.

Amplify through region-specific influencers (Assam and West Bengal) to build buzz and visibility.



RECOMMENDATIONS

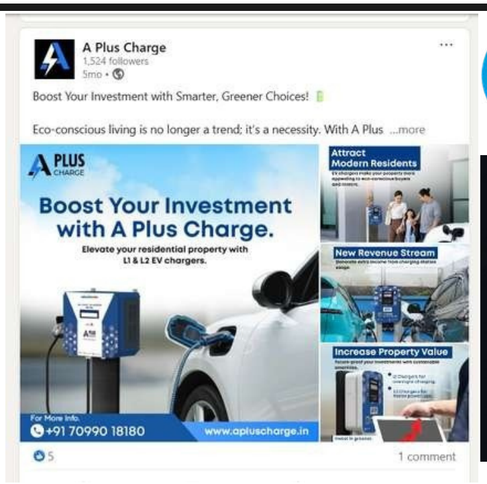
Enhance Social Media — Launch and optimize Instagram, LinkedIn, and YouTube with Reels, Shorts, and professional posts to match 2025 engagement trends.

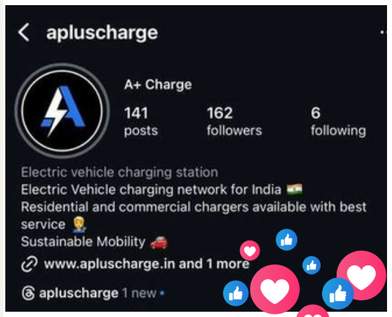
Leverage Social Listening — Use tools like Sprout Social or Brandwatch to track mentions, respond to feedback, and spot trends.

Pursue Media Outreach — Pitch the subscription model's story to tech and automotive outlets (e.g., TechCrunch, Green Car Reports) for coverage.

Optimize Website Content — Add testimonials, blogs, FAQs, and embed social feeds to boost SEO and engagement.

OPTIMIZING A PLUS CHARGE'S ONLINE & SOCIAL MEDIA IMPACT







SOCIAL MEDIA

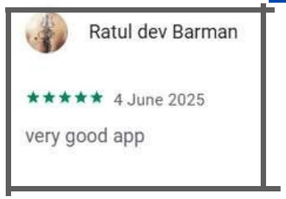
Key Fixes

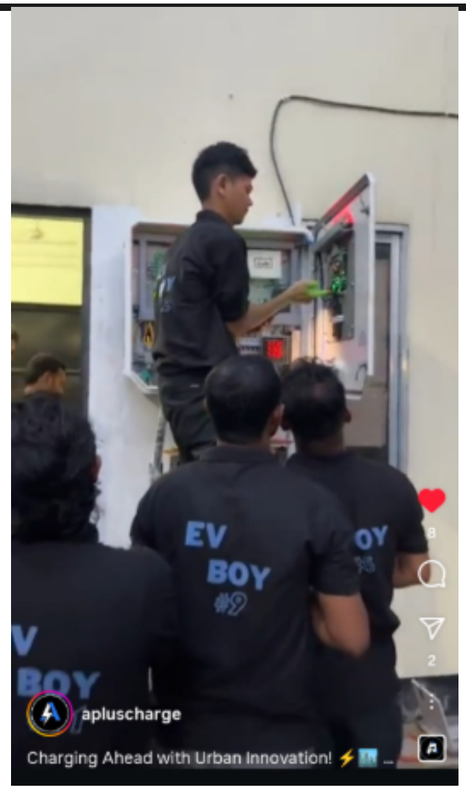
- Add alt-text, hashtags, polls, reels, customer stories → boosts visibility & trust.
- SEO keywords + strong CTAs ("Book a demo") → drives leads & discovery.
- Apply for Blue Tick → improves credibility.
- Revise bio + Linktree + Highlights → better engagement funnel.

Customer Feedback & Reviews

- Complimentary30 kW DC fast charging at Panbazar (GMDA Parking) received praise on Facebook: "Charge your EV absolutelyFREE...Zero Co







Mistakes in the Post

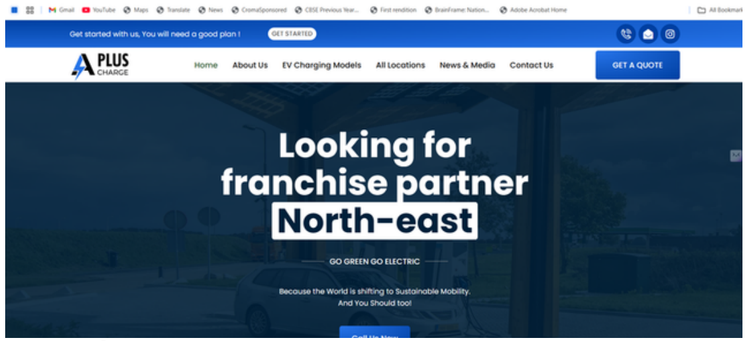
- Posts lack battery progress visuals & context → viewers drop off.
- Website: Weak branding, poor navigation, no trust signals, limited mobile fit.

Our Suggestions

Social Media:

- Loop reels (80%→10%) for instant replays & double watch time.
- Add milestone counters (10%, 30%, 50%, 80%) + hyperlapse full charge to 25 sec.

WEBSITE

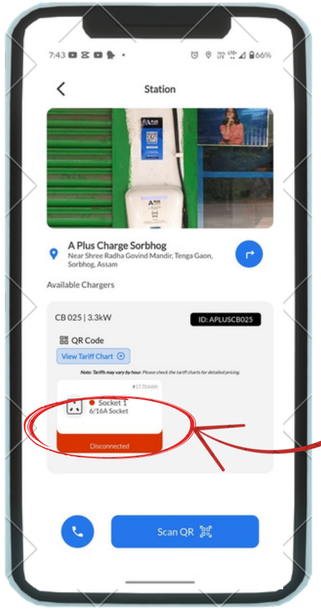


SUGGESTIONS

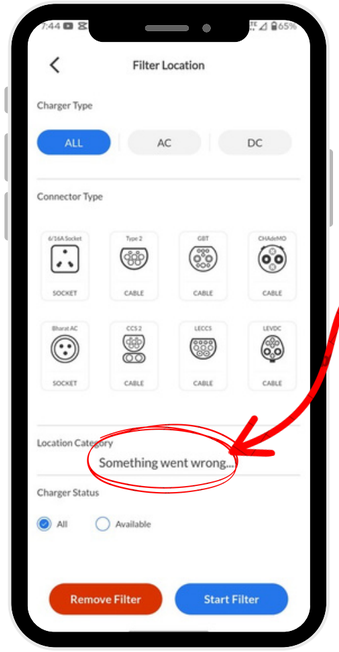
- Strong tagline, clear CTA, interactive map & testimonials for brand building
- Partner logos + mobile-friendly design with ALT text

USER EXPERIENCE DESIGN - SOFTWARE

Current app interface



Step-by-step Instruction
"Could add visual onboarding"

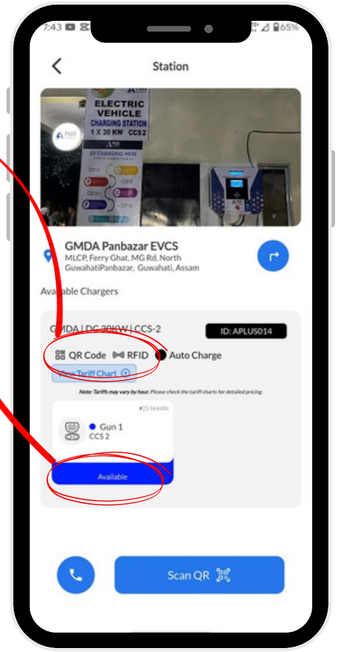


Station Details

- Status ("Available", "Disconnected") unclear; no reason given for disconnection.
- QR/RFID icons without text labels.

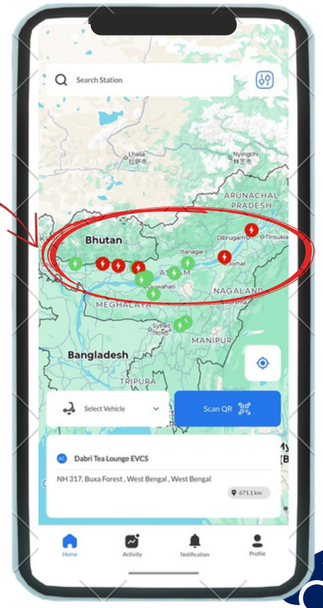
Error Handling

"Location Category" filter returns "Something went wrong...!" with no retry/help.



User Flow

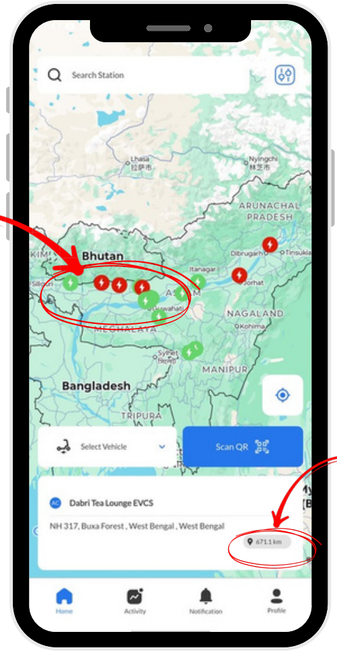
No real-time charging progress updates or completion notifications.



Uneven Color Coding on Map Pins

Navigation

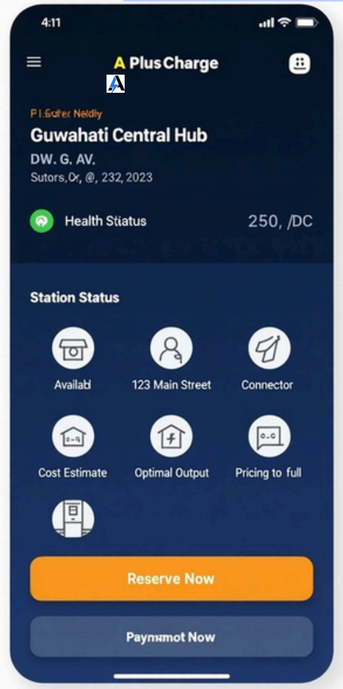
- Map-only view; no list option for scanning stations quickly.
- Charger markers (green/red) have no legend.

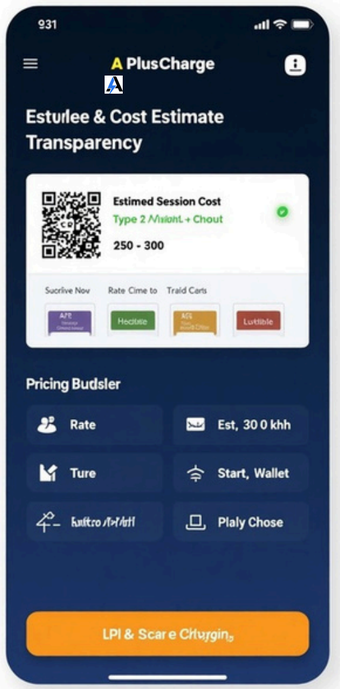


Navigation

Shows faraway stations (600+ km) without sorting/filtering by proximity.

RECOMMENDATIONS





- **Error Handling**: Replace vague messages with clear causes + retry button.
- **Map & Discovery**: Add list view; auto-sort by nearest; add legend for markers.
- **Station Info**: Explain "Disconnected"; show estimated session cost.
- **Start/Stop & Monitoring**: Enable in-app start/stop; show live kWh, cost, time.
- **Payments:** Show available payment methods upfront; allow UPI, card, wallet.
- **Filters**: Show active filters as chips; ensure all categories load reliably.

END-TO-END USER JOURNEY AT APLUSCHARGE

HARDWARE User Journey: EV Driver Using A PlusCharge Charging Station

Discovery & Intent 1

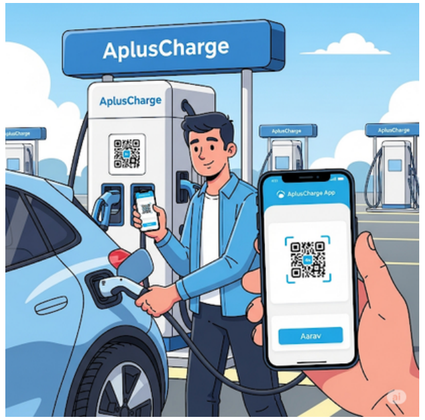


- EV at 20% → Aarav finds APlusCharge on Google Maps.
- Sees real-time availability, charger type, pricing → decides to go.

Key Touchpoint: **Real-time, accurate station info builds trust before arrival.**

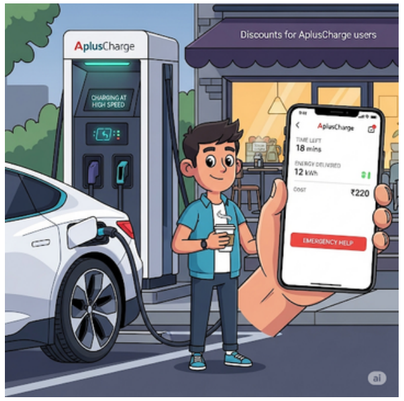
Arrival & Setup 2

- Clear green signage guides him to an open charging bay.
- Authenticates via QR, RFID, or OTP and plugs in within seconds.



Key Touchpoint: **Frictionless start means no queues, no confusion.**

The Charging Experience 3



- App displays remaining time, energy delivered, and total cost.
- Help button available for support, plus partner discounts at nearby outlets.

Key Touchpoint: Live updates + amenities turn downtime into a pleasant break.

Payment & Departure 4



Key Touchpoint: **Effortless checkout leaves a positive final impression.**

Charging stops automatically, and payment is processed instantly via linked UPI.

Digital receipt and CO_2 savings report are emailed, with loyalty points credited to his account.

Feedback & Retention 5

Key Touchpoint: **Timely, personalized nudges keep the user engaged and loyal.**



Leaves a 5★ rating, books his next charging session, and uses a referral offer to earn bonus points.

Receives personalized low-traffic alerts to plan future charges conveniently.

STRATEGIC GROWTH & BUSINESS DEVELOPMENT PLAN

Innovative & Differentiating Solutions

1. Seasonal Green Drives – Special boosts on EV Day, Environment Day, Earth Hour, and festivals (e.g., “Double Coins for Off-Peak Charging this EV Week!”).
2. Rain-Responsive Charging Hood

   Safety + Visibility in Indian weather. A canopy auto-deploys if rainfall is detected by a sensor or the user presses a “cover me” button.
3. Eco-Smart App Interface – Green-toned, minimal-tap design with a live CO_2 saved tracker, making sustainability visible with every charge.
4. Auto-Release Plug – Once charging ends, the head gently self-ejects after 2 minutes, ensuring no “forgotten plug” damage.
5. A+ Coins Rewards – Earn coins for charging, referrals, reviews, and off-peak usage; redeem for free kWh, eco-merch, or partner discounts.
6. Two referral streams: ✅ Host Referral: Existing owners refer new hosts → free installallation/charging days. ✅ User Referral: EV drivers refer friends → free AC/DC minutes.
7. Promoters casually ask in malls/offices “Where do you charge here?” → highlight gap.
   Placards: “This building is EV-Ready with A+ Charge” → triggers neighboring societies to join.
8. Half-Price Tuesdays → predictable mid-week utilization spike. Encourages users to plan charging, improving network stickiness.

Go-To-Market Strategy (GTM)

Targeted expansion into Airbnb, Sports Complexes & Logistics Hubs



Targeted Segment

- Airbnb hosts in NE tourist hubs, sports complexes, logistics corridors.



Localized Value Proposition

- Affordable, reliable charging + “EV-Ready” certification for trust.



Regional Partnerships

- Tie-ups with tourism boards, sports bodies & logistics SMEs.



Distribution & Awareness Channels

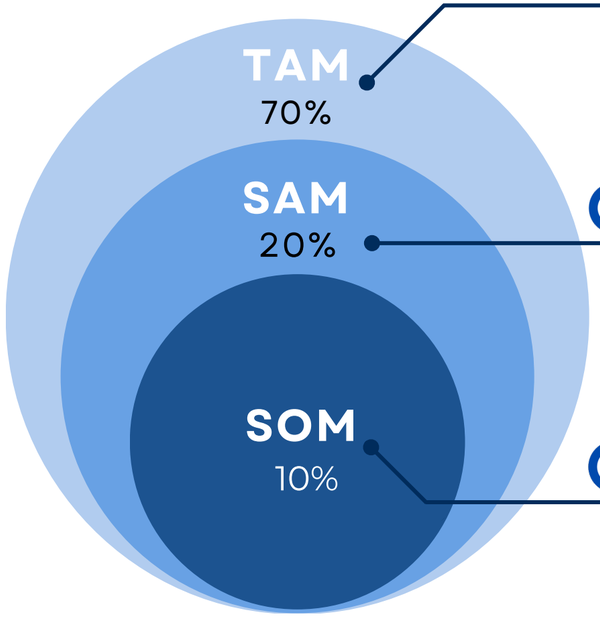
- Digital outreach + grassroots promotion via local influencers & WhatsApp groups.



Cluster-Based Rollout Strategy

- Pilot cities (Guwahati, Siliguri, Kolkata) → expand via franchise & local technicians.

TAM-SAM-SOM + Priority Zones: Where A+ Charge Leads Next



$ 27M

Target for next 2–3 years:

- Launch in 5 cities → 200 host sites × 5 chargers = 1,000 chargers Avg ₹400/session × 100
- sessions/month ₹24L/year/charger
- Total: **~₹240** Cr/year
  = ~USD 25–30 Million

$ 1.7B

Focus:

- Commercial/public charging in top 20 cities (malls, tech parks, co-working spaces)
- Public charging infra expected to capture bulk of near-term demand
- Includes L2/L3 chargers where A+ Charge operates

$ 1.9B

India's entire EV charging infra potential:

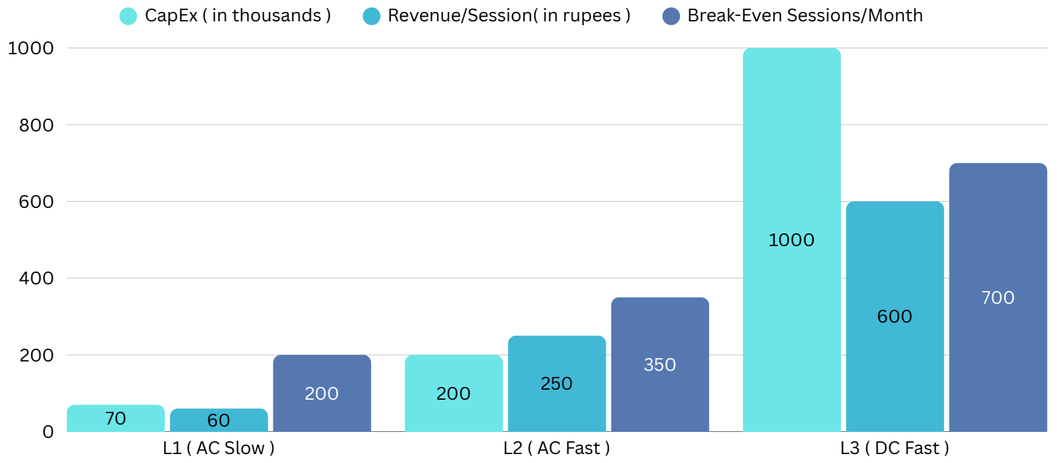
- Market projected to grow from USD 443M (2024) to USD 1.9B by 2030 (CAGR ~28%)
- Includes public, private, and fleet charging across all vehicle types

Target Market

Optimistic 🚀 High EV adoption, clustered network)	**Realistic** ⚖️ (Industry avg, today's utilisation)	**Pessimistic** 🐌 (Slow adoption, poor utilisation)
L1: 300/mo → breakeven **<18m**	**L1:** 150/mo → breakeven **~24m**	**L1:** 60/mo → breakeven **>3y**
L2: 450/mo → breakeven **12–15m**	**L2:** 300/mo → breakeven **~18m**	**L2:** 150/mo → breakeven **~3y**
L3: 600/mo → breakeven **~24m**	**L3:** 360/mo → breakeven **~30m**	**L3:** 180/mo → breakeven **~4–5y**

Upper Range Break-Even by Charger Type (Table/Chart)

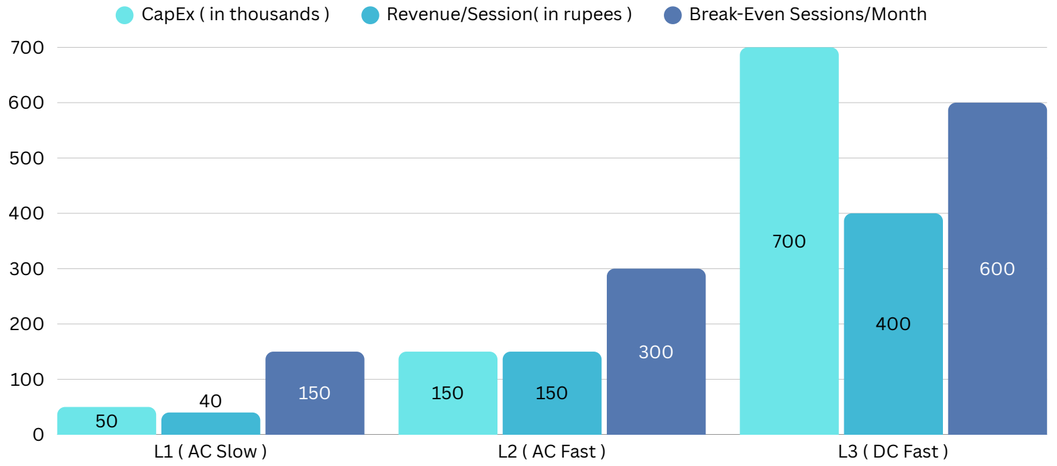
(These numbers are based on OEM/installer data + industry reports + Prevailing tariffs)



AC chargers (7–22 kW) cost ₹50k–75k, DC fast chargers ₹75k–8L in India." — Pulse Energy

Lower Range Break-Even by Charger Type (Table/Chart)

(These numbers are based on OEM/installer data + industry reports + Prevailing tariffs)



AC chargers (7–22 kW) cost ₹50k–75k, DC fast chargers ₹75k–8L in India." — Pulse Energy

RELIABLE GROWTH THROUGH CLUSTERS, PARTNERSHIPS & SMART MODELS



CLUSTER ROLLOUT

- Dense micro-markets (city cores, corridors) → higher utilization.

FRANCHISE MODE

- Local partners share CapEx/Opex → low-cost scaling.

FLEET-FIRST NODES

- Partner with Ola, Amazon, Zipp → guaranteed baseline demand.

ANCHOR PARTNERSHIPS

- Tie-ups with OEMs, malls, fuel stations → premium locations & trust.

CHALLENGES AND LEARNINGS









GLIMPSES OF REALITY:
CONVERSATIONS WITH EV RIDERS

User Inputs

- Majority of EV users in Rourkela rely on free public stations or home charging, with very limited 4W/3W infrastructure.
- Fast charging (~45 min) is strongly preferred over regular (~5 hrs).
- Users demand more frequent, accessible, and reliable public chargers.
- Pricing awareness is low due to free charging; users want transparent, predictable pricing.
- Feedback was mostly verbal/written; many were camera-shy or privacy-conscious.
- Common app issues (esp. Ather app): pairing failures, disconnects, slow loading, location glitches, monthly resets, instability.

Strategic Solution

- Free Charging ≠ Real Behavior – Free usage skews willingness to pay → need gradual shift (free → subsidized → paid).
- Infra Gaps Shape Adoption – Limited 3W/4W charging forces home charging → growth depends on localized infra.
- Low Pricing Awareness – Users unclear on costs → simple, transparent pricing + petrol/diesel comparisons needed.
- Human Trust Matters – Privacy concerns show EV adoption needs comfort + respectful, humanized support.
- Ecosystem Lock-In – Brand loyalty (e.g., Ather) limits feedback → opportunity for universal, multi-brand platforms.

FINANCIAL METRICS - APLUS CHARGE

https://docs.google.com/document/d/18tfn-tsuiCqyh6gJZkNvukFtY6xV6Co6ahQOpNi83Ac/edit?usp=sharing

A PLUS CHARGE

Thank You

IGNITE EDGE

